

WELLSENSE · CALCULATION

WellSense sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (range and resolution, loop supply and energy, accuracy and drift, fit in the well and hanging loads, storage and airtime, vent desiccant, installation, cost)

WellSense sizing calculations

On paper, WellSense meets seven of its sixteen requirements, has four at risk, misses two and leaves three that only field work can settle. The two misses are cost and water safety. With the decisions of WLS-DDR-002 applied (budget raised to \$190, a 22 mm probe in the 1 in tube, a galvanized conduit over the surface cable and a 30 min uplink rule for pumping tests on public networks), the WellSense parts cost \$197.60 at the 30 m design depth, \$7.60 over the \$190 `budget_usd` ; the \$10.00 conduit is the whole overrun, and the budget holds to a probe depth of 25.8 m. No drinking water certificate is in hand for the low-cost cable, seal or probe (R9). The calculations changed the interface: the FieldNode 12 V rail cannot drive a 4 to 20 mA loop through a 150 Ω shunt and 62 m of cable to the transducer's 12 V minimum (it delivers 7.56 V), so the interface board gains a 24 V boost. That doubles the sensor energy of the TRL 2 estimate to 38.2 mWh a day, still only 1.6 % of the FieldNode allowance. Accuracy (R4) is at risk rather than not met: a 0.25 % class probe after a two-point field calibration against a manual tape gives 12.5 mm by root sum square against the ± 20 mm target, but 27.1 mm if every term adds. Every number in this note is printed by `docs/04-calcs/sizing.py` ; the tag in brackets, for example [B3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that anything lowered into a well is safe for drinking water, that the wellhead is sealed, or that work at a well with a mains-powered pump is safe. See WLS-PRC-001, Safety.

■ Scope and method

The note checks every requirement in WLS-REQ-001 v0.4 against the design in WLS-PRC-001 v0.4 and the parametric model `cad/src/model.py` . The script imports the model's `PARAMS` , `DESIGN` and `derived()` , so the tube, probe, casing, post and seal plate used here are the ones in the STEP files and in drawing WLS-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml` . Run it from the repo root with `python docs/04-calcs/sizing.py` ; it also writes `docs/04-calcs/results.csv` .

The design case is a 150 mm (6 in) borehole with a 4 in submersible pump on a 1-1/4 in riser, the probe hung 30 m below the top of the casing (up to 60 m for R2) in a 1 in PVC access tube, and a FieldNode core on a post 0.75 m from the well, reading every 15 min (DDR-001, D5).

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Pressure	Water 1,000 kg/m ³ nominal; standard gravity 9.80665 m/s ² in the sensor's kPa calibration; local gravity from the WGS 84 formula	Physical constants
Loop	150 Ω shunt; ADS1115 class 16-bit ADC at ± 4.096 V; loop ceiling 22 mA; rail on 2 s per reading; FieldNode 12 V rail 5 % low, 90 % efficient; boost to 24 V at 85 %; transducer minimum supply 12 V; two 0.2 mm ² copper cores at 0.087 Ω /m; 0.3 V reverse-polarity diode	BOM lines 1, 2 and 7; FND-CAL-001 for the rail
Node	Controller awake 2 s at 8 mA, 3.3 V, per reading; published sensor allowance 100 mW; 32-byte stored record; 16 MB flash; 20-byte uplink	FND-CAL-001 v0.2
Accuracy	After a two-point calibration: nonlinearity, hysteresis and repeatability 0.10 % of full scale (0.25 % class) or 0.20 % (0.5 % class); probe thermal effect 0.02 % of full scale per K over 2 K of groundwater change; shunt 10 ppm/K and ADC gain 7 ppm/K over 30 K in the junction box; tape reading ± 3 mm (0.01 ft)	Typical datasheet ranges and USGS tape practice; not measured
Mechanical	Vented cable 0.055 kg/m, probe 0.25 kg, strain member 400 N; PVC 1,400 kg/m ³ and 48 MPa tensile	Typical catalogue values
Fit	3 mm running clearance; 2 mm radial clearance for a probe in its tube; riser coupling OD 40 mm (1 in riser in 100 mm casing), 52 mm (1-1/4 in, in 125 and 150 mm casing), 73 mm (2 in, in 200 mm casing); tube coupling OD 40 mm (1 in) and 50 mm (1-1/4 in)	Common pipe sizes
Desiccant	1.4 L of air in the junction box, 1.5 mm vent capillary, 30 K daily swing, 24 g/m ³ humid air, 10 g of silica gel with 20 % usable uptake	Screening values

■ A. Range and resolution (R1, R3)

- **Range.** Full scale is 10 m of water, 98.07 kPa; 1 kPa is 102.0 mm of water [A1]. R1 is met by the 0 to 10 m transducer (DDR-001, D3).
- **Resolution.** The 150 Ω shunt turns 4 to 20 mA into 0.60 to 3.00 V. At 125 μ V per ADC step that is 19,200 steps, or 0.52 mm per step over 10 m, and 1.04 mm on the 0 to 20 m variant [A2]. R3 is met; the 20 m variant just misses 1 mm.
- **Placement.** With the probe at least 1 m below the lowest pumping level, seasonal swing plus pumping drawdown of up to 9 m stays in range with 1 m of headroom; larger swings need the 0 to 20 m variant [A3].

■ B. Loop supply and energy (R8)

- **The 12 V rail cannot drive the loop.** At 22 mA through 62 m of cable (10.8 Ω) and the shunt, the transducer gets 7.56 V from a rail 5 % low, 4.44 V short of its 12 V minimum. A 24 V boost on the interface board gives it 20.16 V, a margin of 8.16 V [B1]. The boost and a 3.3 V regulator for the ADC and barometric sensor are added to BOM line 7 (+\$2).
- **ADC protection.** A 25 mA fault puts 3.75 V on the shunt, above the ADC's 3.6 V limit at a 3.3 V supply, so the ADC input needs a series resistor [B1b].
- **Energy.** Each reading takes 1.38 J in the loop (24 V at 22 mA for 2 s, through the boost and the FieldNode rail) plus 0.053 J for the controller, 1.43 J in all [B2]. At 15 min that is 38.2 mWh a day, 1.59 mW on average: 1.6 % of FieldNode's published 100 mW allowance [B3]. **R8 is met.** During a pumping test at 1 min the load is 0.573 Wh a

day, 23.9 mW, or 24 % of the allowance [B4]. The TRL 2 estimate, 0.66 J and 18 mWh a day, assumed the loop ran straight from the 12 V rail [B5].

■ C. Accuracy and drift (R4, R5)

- **Gravity and density.** Converting kPa to head with standard gravity misreads 10 m by +27 mm at the equator, so firmware uses local gravity and the calibration absorbs the rest [C1]. Water density falls 0.29 % from 4 to 25 °C, 29 mm at 10 m [C2]; groundwater temperature at one well changes little, so firmware uses a fixed site density set at installation.
- **Calibration method.** At installation the caretaker reads depth to water with the tape, then lifts the probe a measured 1 m in its tube and reads again. The two points fix offset and span, removing the datasheet offset and span error, the gravity error and the cable length error. What remains is in Table 2.

Table 2. Error budget at full scale after a two-point calibration, 0.25 % class probe [C3.1] to [C4].

Term	Error
Nonlinearity, hysteresis and repeatability	10.0 mm
Probe thermal effect, 2 K	4.0 mm
Density at a fixed site value, 2 K	3.2 mm
Shunt drift, 30 K	3.8 mm
ADC gain drift, 30 K	2.6 mm
ADC resolution	0.5 mm
Manual tape reference	3.0 mm
Root sum square	12.5 mm
Worst-case sum	27.1 mm

- **R4 is at risk.** The root sum square meets ± 20 mm; the worst-case sum does not, and the nonlinearity figure, the largest term, is a typical value rather than a measured one [C4]. A 0.5 % class probe gives 21.4 mm root sum square and 37.1 mm worst case, so it does not meet R4 even after calibration [C5]; BOM line 1 is therefore specified in the 0.25 % class. Without field calibration the datasheet figures are 50 mm and 25 mm [C6]. An absolute sensor without barometric compensation would read weather as water level, ± 306 mm for ± 3 kPa [C7], which is why the vented design was adopted (DDR-001, D2).
- **R5 cannot be verified at TRL 3.** A quarterly tape check leaves at most 5 mm of a 20 mm-a-year drift uncorrected between checks, close to the 3 mm tape uncertainty [C8]. The real drift of low-cost probes is unknown.

■ D. Fit in the well and hanging loads (R2, R10)

- **Model.** In the 150 mm design well the tube sits 32.2 mm clear of the riser and 18.3 mm from the casing wall. Under WLS-DDR-002 the 1 in tube is kept and the probe body is limited to 22 mm, which gives 2.3 mm radial clearance in the 26.6 mm bore of 1 in Sch 40 PVC [D1], against 1.3 mm for the 24 mm probe of v0.1. Joints must be flush inside so a solvent bead does not catch the probe.

- **Tube size.** The 1 in tube takes probes up to 22 mm with 2 mm of radial clearance and fits beside a centered riser in 150 and 200 mm casings, and beside a riser against the wall in all four sizes checked. A 1-1/4 in tube takes probes up to 28 mm but fits beside a centered riser only in 200 mm casing [D2]. The tube ends above the pump, so the pump body does not limit the fit.
- **R10 is at risk.** The tube and a 22 mm probe fit the design well, but lowering a tube past the riser, cable ties and pump cable of an installed pump may snag them or need the pump pulled. Only an installation trial can settle it.
- **Loads.** Cable and probe hanging 60 m in air weigh 35 N against a 400 N strain member, a factor of 11.5 [D3]. Sixty metres of tube hanging dry weigh 264 N, a stress of 0.82 MPa against 48 MPa, a factor of 58; the seal plate gland needs a tube clamp to carry the 264 N [D4]. **R2 is met.**

■ E. Storage and airtime (R6, R7)

- **R7 is met.** A year of 15 min readings is 35,040 records of 32 bytes, 1.12 MB, 6.7 % of the FieldNode's 16 MB flash [E1]. A 7-day pumping test at 1 min adds 0.32 MB [E2].
- **R6 is met.** The 15 min default and the 1 min test rate are firmware settings. At 1 min the readings are logged and sent 15 at a time in the normal 15 min uplink. That 38-byte batch takes 328.7 ms at SF9, 31.6 s a day, slightly over the 30 s a day fair use of a public network; on a private TwinKit gateway only the regional duty cycle applies [E3]. Under WLS-DDR-002, a test on a public network sends every 30 min instead: a 30-reading batch of 68 bytes takes 472.1 ms, 22.7 s a day, within fair use [E4]. On a private TwinKit gateway the 15 min batch is kept.

■ F. Vent desiccant (maintenance)

The vent capillary holds 110 mL at 62 m. With the box and vent breathing 0.15 L a day at a 30 K swing, humid air carries at most 3.6 mg of water a day; 10 g of gel would last about 552 days even if the whole box breathed through it, a factor of 6.1 over the quarterly change [F1]. The quarterly desiccant change has ample margin, provided the box gaskets are intact.

■ G. Installation (R11)

With an access tube already in place, nine steps from isolating the pump to confirming an uplink are estimated at 120 min for two people, exactly the R11 limit, with the post footing set and cured on an earlier visit [G1]. **R11 cannot be verified at TRL 3; only a timed installation can.**

■ H. Cost (R12)

- **BOM.** All 14 lines are priced. The design case uses 32 m of cable and 30.61 m of tube (31 m in the BOM) [H1].
- **R12 is not met.** Lines 1 to 7, 9, 10, 13 and 14 total \$197.60 against the \$190 `budget_usd` (raised from \$180 under WLS-DDR-002), \$7.60 over; with the \$126.00 FieldNode core the complete logger is \$323.60, and \$331.60 at a hot site with FieldNode's \$8.00 sun shield [H2]. Without the \$10.00 conduit (line 14, also decided under WLS-DDR-002) the parts would be \$187.60, within budget [H2b]. Cable and tube add \$1.80 per metre of depth, so the budget holds to a probe depth of 25.8 m; at 60 m the parts cost \$251.60, and without an access tube (no pump in the casing) \$166.60 [H3]. The combined effect of the \$190 budget and the conduit is raised in `docs/REVIEW.md` as a new item awaiting Amish. Against the TRL 2 estimate of about \$170, the 0.25 % class probe (+\$5), the boost (+\$2), the longer post (+\$2), the concrete (+\$6) and 2 m of surface cable and 1 m more tube (+\$2.60) account for the difference up to v0.1; the conduit (+\$10.00) is added in v0.2.

■ L. Results against every requirement

Table 3. Requirement status from this note [L].

ID	Requirement	Value	Target	Status
R9	Water safety	No drinking water certificate in hand for low-cost cable, seal or probe	Wetted parts 316 stainless or certified for drinking water; well sealed	Not met (no evidence)
R12	Cost	\$197.60 at 30 m; met to 25.8 m; \$323.60 with FieldNode	WellSense parts \$190 or less per well at 30 m	Not met
R4	Accuracy after field calibration	12.5 mm RSS, 27.1 mm worst case (0.25 % class)	±20 mm	At risk
R10	Pump compatibility	2.3 mm radial clearance with a 22 mm probe; tube may need the pump pulled	Own tube, no contact with pump, riser or cable	At risk
R15	Environment	Probe IP68 on datasheet; FieldNode shield at hot sites gives 48.5 to 52.2 °C inside at 45 °C; FieldNode rated to 45 °C ambient	Above-ground parts IP65, -10 to 55 °C; FieldNode shield at hot sites	At risk
R16	Tamper resistance	Cable in galvanized conduit from the tube cap to the box; locking of the wellhead parts not yet specified	Lockable; no exposed cable at reachable height	At risk
R5	Long-term drift	Unknown; a quarterly check leaves at most 5 mm	20 mm a year or less	Not verifiable at TRL 3
R11	Installation	120 min estimate	2 h or less with a tube in place	Not verifiable at TRL 3
R13	Data use	Dashboard not started	Level, drawdown, year-on-year, CSV	Not verifiable at TRL 3
R1	Range	0 to 10 m, 98.07 kPa	0 to 10 m; 0 to 20 m option	Met
R2	Installation depth	Cable factor 11.5, tube factor 58 at 60 m	60 m or more	Met
R3	Resolution	0.52 mm	1 mm or better	Met
R6	Reading interval	15 min; 1 min logged, sent every 30 min on a public network (22.7 s/day)	15 min; 1 min in tests	Met
R7	Local storage	1.12 MB a year, 6.7 % of flash	1 year	Met
R8	Energy	38.2 mWh/day, 1.6 % of 100 mW	5 % or less	Met
R14	Data ownership	Levels, time and well ID only	Community controls sharing	Met (by design)

Counts: 2 not met, 4 at risk, 3 not verifiable at TRL 3, 7 met.

■ Checks against the TRL 2 figures

TRL 2 claim (WLS-PRC-001 v0.2)	This note	Action
Resolution about 0.5 mm	0.52 mm	Stands
Datasheet accuracy about ± 50 mm	50 mm (0.5 %), 25 mm (0.25 %); 12.5 mm RSS after calibration	Precis and REQ updated; R4 now at risk
Energy about 0.66 J per reading, 18 mWh/day	1.43 J and 38.2 mWh/day with the 24 V boost	Precis updated
About 0.6 % of a 2.8 Wh/day allowance	1.6 % of 100 mW (2.4 Wh/day)	Precis updated
Storage about 0.3 MB a year at 8 bytes	1.12 MB at FieldNode's 32-byte record	Precis and REQ updated
Parts about \$170; about \$296 with FieldNode	\$187.60 and \$313.60 in v0.1; \$197.60 and \$323.60 in v0.2	Precis, README and REQ updated
Depth-dependent about \$1.80 per metre	\$1.80 per metre	Stands
12 V rail powers the loop directly	Short by 4.44 V; 24 V boost needed	Precis and BOM updated

■ Changes in v0.2 (WLS-DDR-002)

Item	v0.1	v0.2
<code>budget_usd</code>	\$180	\$190 [H2]
Probe body and clearance	24 mm, 1.3 mm radial	22 mm maximum, 2.3 mm radial [D1]
Surface cable	Exposed from the tube cap to the box	Galvanized conduit, BOM line 14, \$10.00
WellSense parts at 30 m	\$187.60 (\$7.60 over \$180)	\$197.60 (\$7.60 over \$190) [H2]
Pumping test uplinks, public network	15 min batches, 31.6 s/day	30 min batches, 22.7 s/day [E4]
R15 hot sites	FieldNode 45 °C rating only	FieldNode shield required at hot sites (FND-DDR-002)